



Effectiveness of OGAWA Massage Chair on Bone Mineral Density and Biochemical Markers in Post-Menopausal Women: A Randomized Single-Blind Study

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Introduction

Osteoporosis is a major public health concern affecting post-menopausal women, characterized by reduced bone mass and increased fracture risk. Physical interventions such as vibration therapy have shown promise in mitigating bone loss. This study aim to investigate the effect of OGAWA massage chair use on bone mineral density (BMD) and biochemical markers of bone turnover in post-menopausal women.

Method

A randomized single-blind study was conducted involving 42 post-menopausal women, divided into intervention (massage chair) and control groups. Participants were monitored over a 6-month period. BMD was assessed using dual-energy X-ray absorptiometry (DEXA), and bone turnover was evaluated using serum levels of P1NP (formation marker) and CTX-1 (resorption marker).



Figure 1: Patient on a massage chair for the massage session as for mechanical stimuli (vibration and pressure) generated by the massage chair can modulate bone turnover markers

Results

The intervention group showed a significant increase in P1NP from baseline (69.65 ng/mL) to 1 month (75.42 ng/mL), with a slight decrease at 6 months (72.42 ng/mL). CTX-1 remained stable throughout. In contrast, the control group showed a modest increase in CTX-1 levels from baseline (0.464 ng/mL) to 6 months (0.5545 ng/mL) and a less marked increase in P1NP.

The findings suggest that massage therapy via OGAWA massage chairs may promote bone health in post-menopausal women by stimulating bone formation and reducing bone resorption. The mechanical stimulation may mimic low-amplitude vibrations known to enhance osteoblast activity and suppress osteoclast-mediated bone degradation.

These results are consistent with studies on vibration therapy and manual massage effects on bone turnover. While the study was limited by sample size and blinding constraints, the biochemical trends observed support further investigation into massage-based interventions for osteoporosis prevention.

Conclusions

Regular use of the OGAWA massage chair showed a favorable impact on bone turnover markers, supporting its role as a complementary non-pharmacologic strategy in post-menopausal bone health management. Larger trials with longer follow-up are warranted.

References

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